

SQL Joins Master Reference Guide

Comprehensive Technical Notes: Inner, Left, Right & Full Outer Joins

Relational Databases

MySQL / PostgreSQL

Power BI Integration

Core Concept: SQL Joins combine rows from two or more tables based on a related column between them (Foreign Key → Primary Key). Understanding joins is essential for combining schema tables (like Employee and Projects) before modeling in Power BI or analyzing relational databases.

1. Sample Dataset Context

Throughout these notes, we refer to the following relational structure from your database model:

Employee Table (Left Table)

Id	FName	Department
1	John	IT
2	James	Finance
3	Mary	HR
4	Alex	Analytics

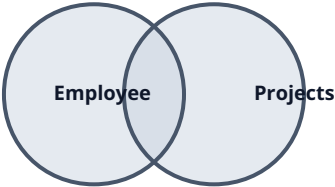
Projects Table (Right Table)

ProjectId	ProjectName	EmployeeId
101	Hosting Fix	1
102	AI Ecommerce	1
103	Portal Redesign	3
104	Cloud Migration	NULL

2. Visual Breakdown of Join Types

A. INNER JOIN (Exact Matches Only)

Returns only the records that have matching values in **both** tables. Unmatched rows from either side are completely excluded.



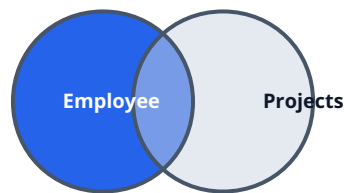
```
SELECT e.Id, e.FName, e.Department, p.ProjectName
FROM Employee e
INNER JOIN Projects p ON e.Id = p.EmployeeId;
```

Resulting Output:

Id	FName	Department	ProjectName
1	John	IT	Hosting Fix
1	John	IT	AI Ecommerce
3	Mary	HR	Portal Redesign

B. LEFT JOIN (All Left + Matching Right)

Returns **all records from the left table** (Employee), and the matched records from the right table (Projects). If there is no match, NULL values appear for right table columns.



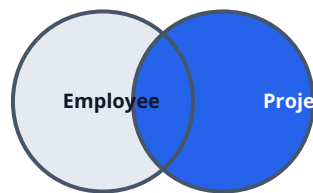
```
SELECT e.Id, e.FName, e.Department, p.ProjectName
FROM Employee e
LEFT JOIN Projects p ON e.Id = p.EmployeeId;
```

Resulting Output:

Id	FName	Department	ProjectName
1	John	IT	Hosting Fix
1	John	IT	AI Ecommerce
2	James	Finance	NULL
3	Mary	HR	Portal Redesign
4	Alex	Analytics	NULL

C. RIGHT JOIN (All Right + Matching Left)

Returns **all records from the right table** (Projects), and matched records from the left table (Employee).
Unassigned projects display NULL for employee details.



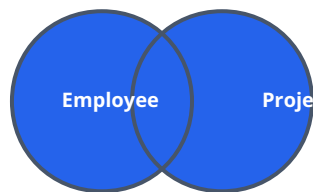
```
SELECT e.FName, p.ProjectId, p.ProjectName
FROM Employee e
RIGHT JOIN Projects p ON e.Id = p.EmployeeId;
```

Resulting Output:

FName	ProjectId	ProjectName
John	101	Hosting Fix
John	102	AI Ecommerce
Mary	103	Portal Redesign
NULL	104	Cloud Migration

D. FULL OUTER JOIN (Complete Union of Both)

Returns **all records when there is a match in either left or right table**. It combines the results of both LEFT JOIN and RIGHT JOIN.



```
SELECT e.FName, p.ProjectName
FROM Employee e
FULL OUTER JOIN Projects p ON e.Id = p.EmployeeId;
```

MySQL Syntax Note: MySQL does not directly support FULL OUTER JOIN. In MySQL, achieve a Full Outer Join by combining LEFT JOIN and RIGHT JOIN using UNION:

```
SELECT e.FName, p.ProjectName FROM Employee e LEFT JOIN Projects p ON e.Id =
p.EmployeeId
UNION
SELECT e.FName, p.ProjectName FROM Employee e RIGHT JOIN Projects p ON e.Id =
p.EmployeeId;
```

3. Executive Summary Matrix

Join Type	Included Rows	Use Case / Analytical Purpose
INNER JOIN	Only matching rows in both tables	Filtering active employees who are currently assigned to projects.
LEFT JOIN	All left rows + matching right rows	Auditing employee workload (identifying employees with 0 projects).
RIGHT JOIN	All right rows + matching left rows	Auditing unassigned/orphaned projects requiring resource allocation.
FULL OUTER JOIN	All rows from both tables	Complete operational audit across entire corporate inventory.